SWAPS VS. FORWARDS

Learning Outcome

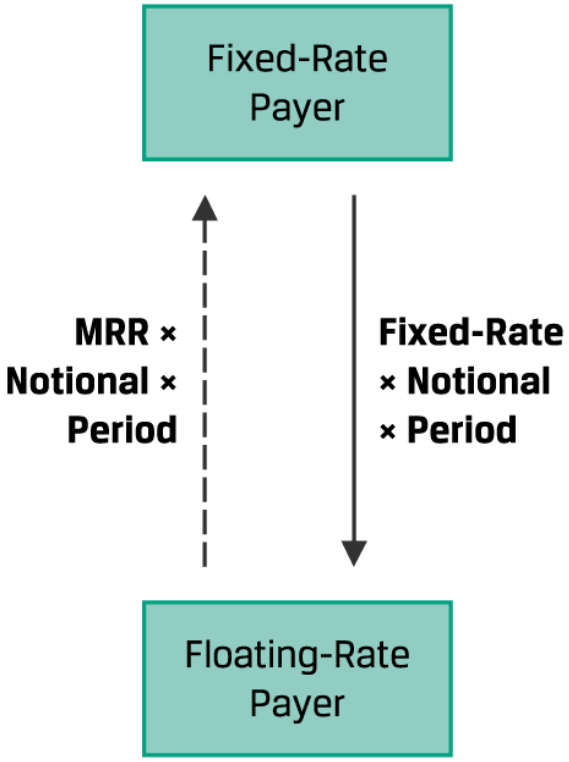
- describe how swap contracts are similar to but different from a series of forward contracts

A swap contract is an agreement between two parties to exchange a *series* of future cash flows, while a forward contract is an agreement for a *single* exchange of value at a later date. Although this lesson focuses on interest rate swaps, similar principles apply to other underlying variables where a series of cash flows are exchanged on a future date.

An earlier lesson showed how implied forward rates may be derived from spot rates. An implied forward rate for a given period in the future is equivalent to the forward rate agreement (FRA) fixed rate for that same period for which no riskless profit opportunities exist. The single cash flow of an FRA is similar to a single-period swap, as shown in Exhibit 1.

Exhibit 1:

Swap and FRA Payoff Profile



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In each case, the net difference between a fixed rate agreed on at inception and an MRR set in the future is used as the basis for determining cash settlement on a given notional principal over a specific time period. For example, a fixed-rate payer on a swap or FRA will realize a gain if the MRR sets at a rate higher than the agreed-on fixed rate and will receive a net payment from the floating-rate payer. However, as we saw in an earlier lesson, the FRA has a single settlement, which occurs at the *beginning* of an interest

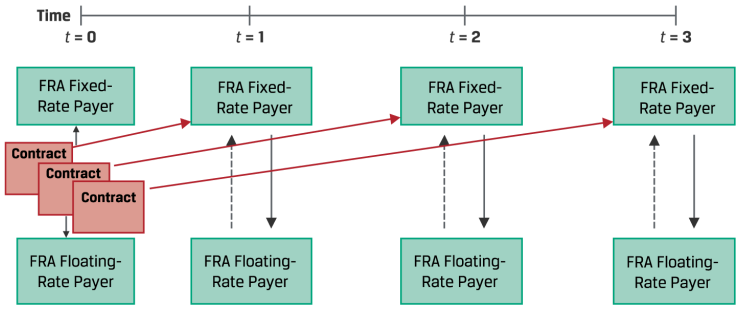
Other similarities between interest rate forwards and swaps include the symmetric payoff profile and the fact that no cash flow is exchanged upfront. Both interest rate forward and swap contracts involve counterparty credit exposure.

Since interest rates are characterized by a term structure, different FRA fixed rates usually exist for different times to maturity. In contrast, a standard interest rate swap has a *constant* fixed rate over its life, which includes multiple periods. This relationship is shown visually in Exhibit 2 and numerically in Example 1, which extends an earlier spot and forward rate example.

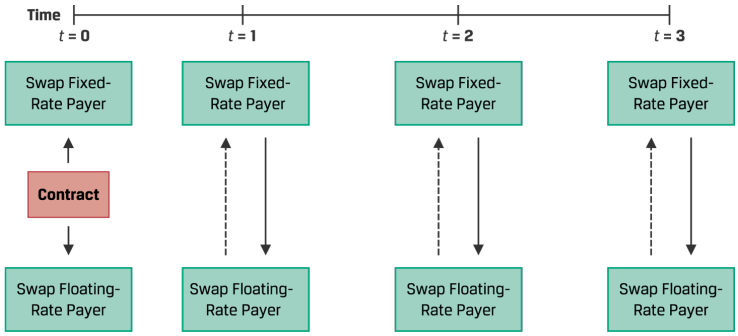
Exhibit 2:

Series of FRAs vs. Standard Interest Rate Swap

Series of Forward Rate Agreements (at Different Fixed Rates)



Standard Interest Rate Swap (At A Constant Fixed Rate)



EXAMPLE 1

Recall from an earlier lesson that three recently issued annual fixed-coupon government bonds had the following coupons, prices, yields-to-maturity, and zero (or spot) rates:

Years to Maturity	Annual Coupon	PV (per 100 FV)	YTM	Zero Rates
1	1.50%	99.125	2.3960%	2.3960%
2	2.50%	98.275	3.4068%	3.4197%
3	3.25%	98.000	3.9703%	4.0005%

We solve for the implied forward rate ($IFR_{A,B-A}$), the break-even reinvestment rate for a period starting in the future (at $t = A$) between short-dated (z_A) and longer-dated (z_B) zero rate using the following formula:

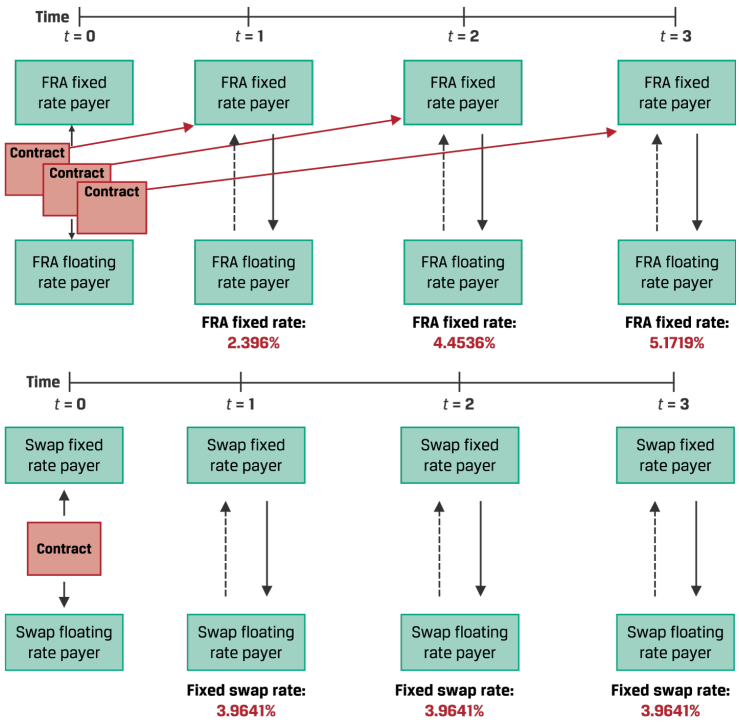
$$(1 + z_A)^A \times (1 + IFR_{A,B-A})^{B-A} = (1 + z_B)^B. \quad 1$$

$IFR_{0,1}$ = 2.3960% = (1.023960) – 1.

$IFR_{1,1}$ = 4.4536% = $(1.034197)^2/(1.02396) - 1$.

$IFR_{2,1}$ = 5.1719% = $(1.040005)^3/(1.034197)^2 - 1$.

One way to create a forward commitment for multiple periods with an initial value of zero would be to use a series of forward rate agreements, exchanging cash flows based on the respective implied forward rate (i.e., the FRA fixed rate) for each period. However, the fixed rate would *vary* for each period based on the term structure of interest rates. Instead, a standard interest rate swap is characterized by a *constant* fixed rate over multiple periods. These rates are shown for both alternatives below.



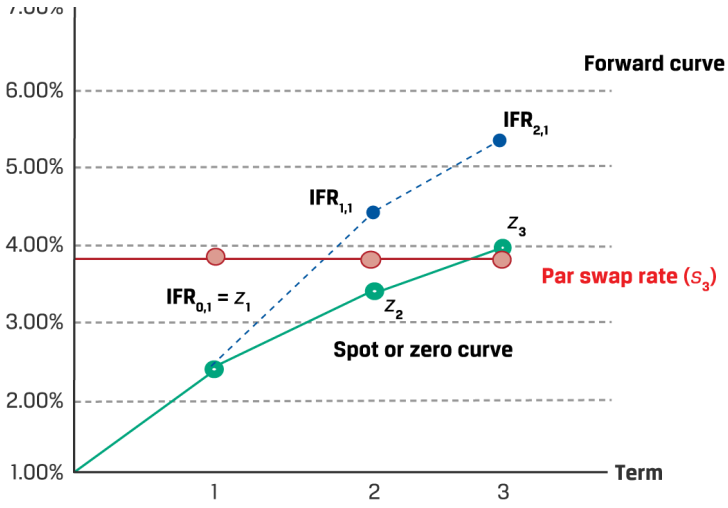
The method used to solve for a swap rate was introduced in an earlier fixed-income lesson. The *par rate* (PMT) was shown to be the fixed rate at which a fixed-coupon bond has a price equal to par (or 100) using a sequence of zero rates (z_i for period i) as market discount factors, as follows:

$$100 = \frac{PMT}{(1+z_1)^1} + \frac{PMT}{(1+z_2)^2} + \cdots + \frac{PMT+100}{(1+z_N)^N}. \qquad 2$$

A standard interest rate swap represents an exchange of fixed payments (with *no* final principal payment) at a constant rate for a series of floating-rate cash flows expected to equal the respective implied forward rates at time t = 0. We must therefore modify the par bond rate calculation in Equation 2 to solve for PMT as a **par swap rate**.

Exhibit 3:

Par Swap Rate, Spot, and Forward Curve



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The par swap rate is the fixed rate that equates the present value of all future expected floating cash flows to the present value of fixed cash flows.

Σ PV(Floating payments) = Σ PV(Fixed payments), or

$$\sum_{i=1}^{N} \frac{\text{IFR}}{(1+z_i)^i} = \sum_{i=1}^{N} \frac{s_i}{(1+z_i)^i}. \qquad 3$$

In our three-period example, we use the implied forward rates, IFR, to solve for s_3, or the three-year swap rate:

$$\frac{\text{IFR}_{0,1}}{(1+z_1)} + \frac{\text{IFR}_{1,1}}{(1+z_2)^2} + \frac{\text{IFR}_{2,1}}{(1+z_3)^3} = \frac{s_3}{(1+z_1)} + \frac{s_3}{(1+z_2)^2} + \frac{s_3}{(1+z_3)^3}.$$

Substitute each of these building blocks into Equation 3 to solve for s_3:

$$0.111017 = \frac{2.396\%}{1.02396} + \frac{4.4537\%}{(1.034197)^2} + \frac{5.1719\%}{(1.040005)^3} = \frac{s_3}{1.02396} + \frac{s_3}{(1.034197)^2} + \frac{s_3}{(1.04000}$$

$0.111017 = 2.800545 \times s_3$.

$s_3 = 3.9641\%$.

The three-year swap rate of 3.9641% may be interpreted as a multiperiod breakeven rate at which an investor would be indifferent to

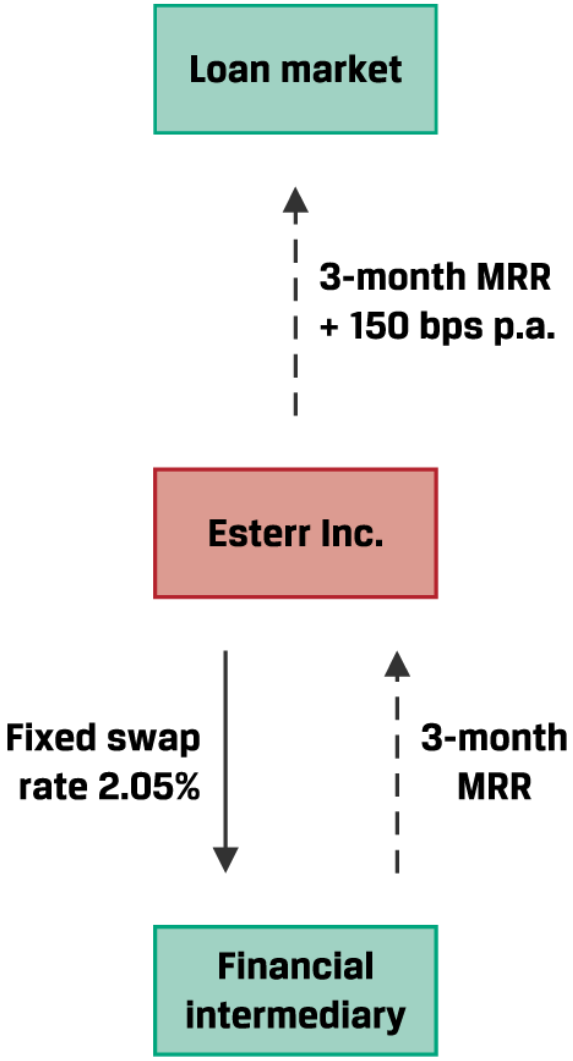
- *paying* the fixed swap rate and *receiving* the respective forward rates or
- *receiving* the fixed swap rate and *paying* the respective forward rates.

For this reason, we may think of the fixed swap rate as an internal rate of return on the implied forward rates through the maturity of the swap as of $t = 0$. Exhibit 3 also demonstrates that while the combined value of the equivalent forward contracts is zero at time $t = 0$, some individual forward exchanges may have a positive present value and some will have a negative present value. For example, a fixed-rate receiver (floating-rate payer) on the swap in Example 1 would expect the following cash flow in one year's time:

- *Receive* the fixed swap rate (s_3) of 3.9641%.
- *Pay* the initial floating rate ($\text{IFR}_{0,1}$) of 2.396%.
- *Receive* a net payment of 1.5681% (= 3.9641% – 2.396%) on the notional for the period.

Derivative end users, such as issuers and investors, tend to use swaps more often than individual interest rate forward contracts for a number of reasons. For example, the ability to match cash flows of underlying assets or liabilities allows issuers to transform their exposure profile as in Example 2.

Recall from an earlier lesson that Esterr Inc. has a CAD250 million floating-rate term loan at three-month Canadian MRR plus 150 basis points with three and a half years to maturity paid quarterly. Esterr enters into a CAD250 million interest rate swap contract to pay a fixed quarterly swap rate of 2.05% and receive a three-month floating MRR on a notional principal of CAD250 million based on payment dates that match the term loan. The combined loan and swap exposure may be shown as follows:



Under the swap arrangement, Esterr can fix its interest expense for the term loan and avoids the administrative burden of multiple forward contracts at different forward rates. Consider the following cash flow scenarios on the interest rate swap and term loan (assuming each interest period is 0.25 year):

Scenario 1:

CAD MRR sets at 3.75%. As the fixed-rate payer, Esterr

- pays the fixed swap rate of 2.05% and receives CAD MRR of 3.75%,
- *receives* a *net* swap payment of 1.70% for the quarter, and
- makes a term loan payment of 5.25% (= 3.75% + 1.50%).

Esterr's interest expense is 3.55% (= 5.25% paid – 1.70% received).

Scenario 2:

CAD MRR sets at 1.25%. As the fixed-rate payer, Esterr

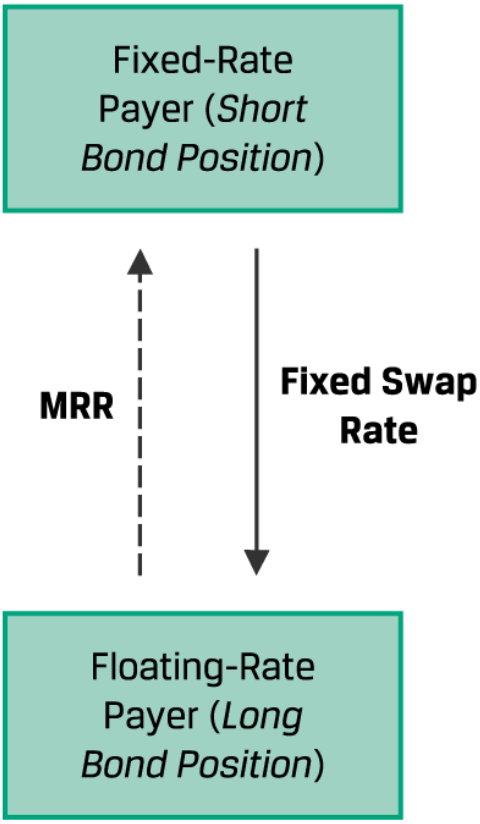
- pays the fixed swap rate of 2.05% and receives CAD MRR of 1.25%,
- *pays* a *net* swap payment of 0.80% for the quarter, and

Regardless of where CAD MRR sets each period, Esterr's interest expense for each quarterly interest period is approximately CAD2,218,750 (= 3.55%, [equal to the 2.05% swap rate + 1.50% loan spread] × CAD250 million × 0.25 year).

As a fixed-income instrument with periodic fixed cash flows through maturity, an interest rate swap should be expected to have risk and return features similar to those of a fixed-coupon bond of a similar maturity. This feature makes interest rate swaps a more attractive alternative to forward rate agreements for investors in managing fixed-income exposures as well. Consider the following comparison of a cash bond position to paying or receiving a fixed swap rate:

Exhibit 4:

Using Swaps to Manage Fixed-Income Exposure



Instrument	Position	Higher interest rates	Lower interest rates
Cash bond	Long fixed bond Short floating-rate note	Loss	Gain
Interest rate swap	Receive fixed Pay floating	Loss	Gain
Cash bond	Short fixed bond Long floating-rate note	Gain	Loss
Interest rate swap	Pay fixed Receive floating	Gain	Loss

While the next lesson will explore the value and price of interest rate swaps in greater detail, Exhibit 4 demonstrates the similarity between a long (short) bond position and a receive (pay) fixed interest rate swap. Given their greater liquidity than and similar benchmark exposure profile to individual bond positions, active fixed-income portfolio managers often use swaps rather than underlying securities to adjust their interest rate

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in a lower rate environment.

As mentioned earlier, FRAs are primarily used by financial intermediaries to manage their rate-sensitive positions on a period-by-period basis. Issuers and investors typically opt for the greater efficiency of interest rate swaps to manage their interest rate exposures. As we will see in later lessons, the greater liquidity of interest rate swaps has also led to their use both as a bond pricing benchmark and an underlying variable for other derivative instruments.

Question Set: Interest Rate Swaps vs. Forward Contracts

1. Determine the correct answers to fill in the blanks: A fixed-rate payer on a swap or FRA will realize a(n) _____ if the MRR sets at a rate higher than the agreed-on fixed rate and will _____ a net payment _____ the floating-rate payer.

Hide Solution

Solution:

A fixed-rate payer on a swap or FRA will realize a *gain* if the MRR sets at a rate higher than the agreed-on fixed rate and will *receive* a net payment *from* the floating-rate payer.

2. Identify which of the following characteristics matches which forward commitment contract.

1. The price of this contract is the implied forward rate, or the breakeven reinvestment rate, for a period starting in the future.	A. Interest rate swap
2. Involves counterparty credit risk	B. Forward rate agreement
3. Has a constant fixed rate for which the present value of future fixed versus floating cash flow exchanges is equal to zero	C. Both an interest rate swap and an interest rate forward contract

Hide Solution

Solution:

1. B is correct. The breakeven reinvestment rate, or implied forward rate, is the no-arbitrage FRA fixed rate.
2. C is correct. Both an interest rate swap and a forward rate agreement involve counterparty credit risk.
3. A is correct. An interest rate swap has a constant fixed rate for which the present value of fixed versus floating cash flow exchanges equals zero.

3. Identify which of the following benefits of using swaps over forwards are most applicable to which derivative end users.

1. Swaps allow these end users to match the periodic cash flows of a specific balance sheet liability to transform their interest rate exposure profile.	A. Both issuers and investors
2. Swaps enable these end users to actively adjust their interest rate exposure profile without buying or selling underlying securities	B. Issuers
3. Swaps involving a series of cash flows enable these end users to avoid the administrative burden of entering into and managing multiple forward contracts.	C. Investors

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Home
Study Plan
Lessons
Flashcards
Practice
Mock Exams
Game Center
Discussions
Search

1. B is correct. Swaps allow issuers to match the periodic cash flows of a specific balance sheet liability to transform their interest rate exposure profile, as shown in Example 2.
2. C is correct. Swaps enable investors to actively adjust their interest rate exposure profile without buying or selling underlying securities.
3. A is correct. Both issuers and investors benefit from a reduced administrative burden of entering one interest rate swap for a series of cash flows rather than multiple individual forward contracts.

4. Identify the following statement as true or false, and justify your response: The market reference rate (MRR) is the internal rate of return on the implied forward rates over the life of an interest rate swap.

Hide Solution

Solution:

False. The fixed swap rate is the internal rate of return on the implied forward rates over the life of an interest rate swap.

5. Explain how an active fixed-income portfolio manager might use an interest rate swap rather than underlying bonds to realize a gain in a lower–interest rate environment, and justify your response.

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Solution:

A manager may choose to receive fixed, pay floating on an interest rate swap, with the fixed cash flow stream being similar to owning a fixed-coupon bond. If interest rates are expected to fall, the manager will realize an MTM gain in a lower-rate environment.

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